

Chapter – 3: Mindful Eating: A Path to a Healthy Body

Who is healthy? Who is healthy? Who is healthy? One who eats food that is wholesome, in moderate quantities, and appropriate for the season, time, and place. – Wise saying

- Medu, Mishti – thought of the day – read daily
- Today's thought – *annena jatani jivanti* – Sanskrit saying – meaning – food gives life to living beings

What Do We Eat?

- Activity –
 - Food – eaten daily – essential component
 - List food items – consumed in a week
- **Compare** – list with friends – find similarities and diff.
- Lots of variety – food eaten – diff. regions

Food in diff. regions

- Activity –
 - Find out – types of foods – traditionally consumed – crops grown – various states
- **Analyse** – data collected – some food items – common – some other – consumed in particular states
- Traditional food – based on – crops grown – particular states
- India – agricultural country – diverse soil, climate – various crops – grown
- Choice of food – vary based on **cultivation** of crops – particular region, taste preferences, culture, tradition

How have cooking practices changed over time?

- Food habits – vary – through states
- Food choices, preparation methods – also vary
- Activity –
 - Prepare list – ques. – gather info. – elder people – food habits
 - Kind of food – still eating – what's new
 - Changes – cooking practices
 - Cause of change
 - Conduct interviews – elderly people – based on ques.
- Cooking practices – also called **culinary practices** – changed
- Imp. diff. – traditional, modern culinary practices
- Earlier –
 - Cooking – *chulha*
 - Grinding – *sil-batta*
- Now a days –
 - Cooking – gas stove
 - Grinding – electrical grinder
- Changes – factors – tech. development, improved transport, better communication

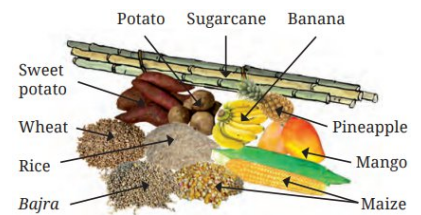
What are the Components of Food?

- Medu, Mishti – traditional food festival – their school – theme – eat healthy, live healthy

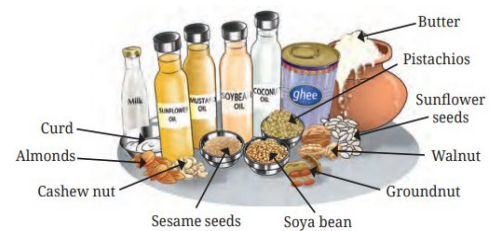
- Festival – various stalls – diff. traditional dishes
- Dr. Poshita – nutrition expert – explains – health is the ultimate wealth
- Miss a meal – feel tired, less energetic

- Marathon runner – drinks glucose – instant energy
- Glucose – **carbohydrate** – primary src. – energy
- Src. of carbs. –

- Cereals – wheat, rice, maize
- Vegetables – potato, sweet potato
- Fruits – banana, pineapple, mango

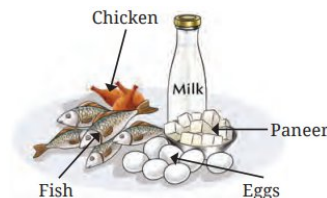
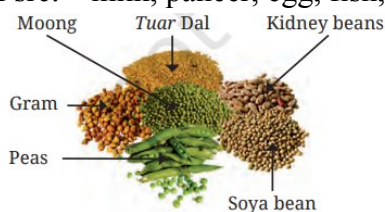


- Common sugar – good src. – carbs.
- *Laddoos* – part of traditional diet – winters
 - *Besan*, wheat flour (*aata*), ghee, *goond* (edible gum), nuts, seeds – main ingredients



- Ghee, various oils – grouped under – **fats**
- Src. – both plants, animals
 - Nuts – groundnuts, walnuts, coconuts, almonds
 - Seeds – pumpkin, sunflower
- Fat – src. – stored energy
- Carbs., fats – provide energy – various act. – **energy-giving foods**

- **Proteins** – imp. part – our food
- Milk products, pulses – good src.
- Sportspersons – need proteins – large qty. – build muscles
- Plant src. – pulses, beans, peas, nuts
- Animal src. – milk, paneer, egg, fish, meat

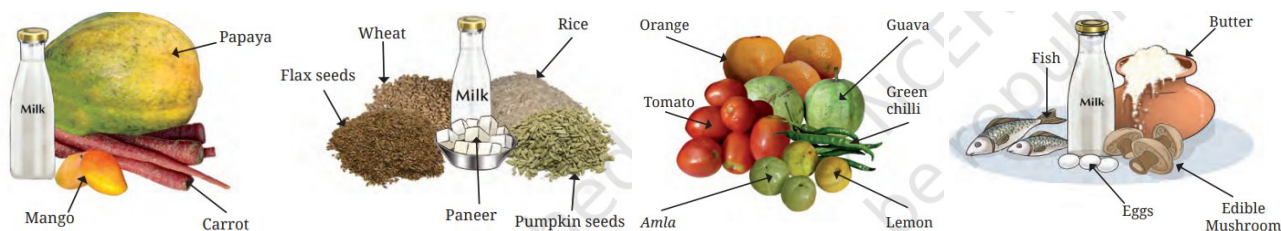


- Protein-rich food – helps – growth, repair of body – **body-building foods**
- Right amt. – protein – children's diet – proper growth, development

- Imp. – **food components** – 2 cases –

- Case 1 –
 - Earlier times – long voyages – sailors – suffered – bleeding, swollen gums
 - Voyage, 1746 – Scottish physician James Lind – obs. – sailors – consumed lemons, oranges – recovered
 - Bleeding, swollen gums – symptoms – **scurvy**
- Scurvy – deficiency – Vitamin C
- Citrus fruits – lemons, oranges – contain **Vitamin C**
- Case 2 –
 - 1960s – humans – Himalayan region, Northern plains – symptoms – swelling of neck
 - Norms – Govt. of India – supplement common salt – iodine – **Iodised salt**
 - Consumption – Iodised salt – reduced – above symptoms – **Goitre**
 - These symptoms – deficiency – iodine – lack of iodine – common diet

Food component	Functions	Some src.	Deficiency disease	Symptoms
Vitamin A	Keep eyes, skin healthy	Papaya, carrot, mango, milk	Loss of vision	Poor vision, night blindness, sometimes complete loss
Vitamin B ₁	Keep heart healthy – help body – perform other func.	Legumes, nuts, whole grains, seeds, milk, products	Beriberi	Swelling, tingling, burning – feet, hands – trouble in breathing
Vitamin C	Helps body – fight disease	<i>Amla</i> , guava, green chilli, orange, lemon	Scurvy	Bleeding gums, slow healing of wounds
Vitamin D	Helps body – absorb calcium	Exposure to sunlight, milk, butter, fish, eggs	Rickets	Soft, bent bones
Calcium	Keeps bones, teeth – healthy	Milk, soya milk, curd, cheese, paneer	Bone, teeth decay	Weak bones, tooth decay
Iodine	Helps – phy., mental act.	Seaweed, water chestnut (<i>singhada</i>), Iodised salt	Goitre	Swelling – front of neck
Iron	Imp. component – blood	Green leafy vegetables, beetroot, pomegranate	Anaemia	Weakness, shortness of breath



- **Vitamins** (A, B₁, C, D), **minerals** (calcium, iodine, iron) – 2 grp. – food components – protect our body – various diseases
- Food components – provide energy, support growth, help repair, protect body, maintain body func. – **nutrients**
- Major nutrients – our food – carbs, proteins, fats, minerals, vitamins
- Vitamins, minerals – **protective nutrients** – protect from disease – keep us healthy
- Parents – advise – milk, green vegetables, fruits, wholegrains – eat regularly
- Food items – src. vitamins, minerals – req. – small amt. – very essential
- Diff. – raw and cooked – some vegies – loose colour – some – loose crisp
- Some nutrients – lost – cooking at high flame
- Washing – cut, peeled – fruits, vegies – loose nutrients
- Fruits, vegies – rich – dietary fibres
- Our body – needs dietary fibres, water
- Fibres – also called – **roughage** – no nutrients – BUT – help body – remove undigested food
- Roughage – provided by – plant products – green leafy vegies, fresh fruits, whole grains, pulses, nuts – good src.
- Eating local food – healthy for body – healthy for env.
- **Water** – imp. part – helps body – absorb nutrient
- Remove waste – sweat, urine
- Drink sufficient water – keep healthy

- **Coluthur Gopalan** – nutrition research – analyzed 500+ Indian food – their nutritional value – recommended diet
- Surveys – nutritional status – Indian population – identify deficiencies – protein, energy, other components
- Implementation – Mid Day Meal, 2002 – provide balanced food – govt.-run, govt.-aided schools

How to Test Different Components of Food?

- Some nutrients – **starch**, fat, protein – detect – simple test
- Others – well-equipped lab

Test for starch

- Activity –
 - Small qty. – food items – potato, cucumber, bread, boiled rice, boiled gram, peanuts, oil, butter, coconut
 - Each item – separate dish – dropper – 2-3 drops – dil. iodine solution
 - Obs. – change of colour
- Blue-black colour – indicate – presence of starch

Test for fats

- Activity –
 - Small qty. – food items – tested above
 - Each food – separate paper – wrap it – press it – do not tear it
 - Food item – contain water – let paper dry
- Oily patch – indicate – presence of fats

Test for proteins

- Activity –
 - Food items – previous activity – make a paste – pestle, mortar
 - ½ teaspoon – each food – separate test tube – add 2-3 teaspoons – water
 - Add 2 drops – copper sulphate solution – dropper – each test tube
 - Another dropper – add 10 drops – caustic soda solution – each test tube
 - Shake well – leave undisturbed – few minutes
 - Obs. – change in colour
- Violet colour – indicate – presence of proteins
- Peanuts – both fats, proteins
- Some foods – multiple nutrients
- Other foods – lack of nutrients

Balanced Diet

- Req. – type, amt. – nutrition – vary – factors – age, gender, phy. act., health status, lifestyle, etc.
- Diet – all nutrients, roughage, water – right amt. – proper growth – **balanced diet**
- Activity –
 - Read – nutritional info. – potato wafers, roasted *chana* – compare
- Some foods – high calories – high sugar, fat content

- These foods – low amt. – proteins, minerals, vitamins, dietary fibres – junk foods
- Ex. – potato wafers, candy bars, carbonated drinks – not healthy
- Eating balanced diet – avoiding junk food – healthy body

Millets: Nutrition-rich Cereals

- *Jowar, bajra, ragi, sanwa* – native crops – India
- Cultivation – easy – diff. climates
- Highly nutritious – called – **millets**
- Small-sized – integral part – Indian diet
- Good src. – vitamins, minerals – iron, calcium – dietary fibres

Food Miles: From Farm to Our Plates

- Entire dist. – any food item – producer to consumer – **food miles**
- Reduce – food miles – reduce cost – reduce pollution – support local farmers
- Many people – waste food – very bad
- People – remember – time, effort – produce food – not waste it

